

Scheduler — Black-box I/O

One channel, DDR5. The scheduler = the final select/pack bar. The per-rank ARBITERS (rank_act_abr, rank_cas_abr, pick-2-pre) sit UPSTREAM and hand it the winners; the scheduler packs the bundle and drives the phaser. **in** = into scheduler, **out** = from scheduler.

1 INPUTS (into the scheduler)

Per-rank arb winners (from rank_act_abr / rank_cas_abr / pick-2-pre)

signal	width	dir	meaning
act_win[r] = {rdy, bg, bank}	few × N_RANKS	in	the 1 ACT winner / rank (already & can_act_rank_s upstream)
act_row[r]	row × N_RANKS	in	row for that ACT (row_sel)
cas_win[r] = {rdy, bg, bank}	few × N_RANKS	in	the 1 CAS winner / rank (& can_cas_rank_s upstream)
cas_col[r] = {idx, wr/rd_ptr, column}	bus × N_RANKS	in	col + sram ptr + rob for that CAS (col_sel)
pre_win[r][0..1] = {rdy, bg, bank}	few × 2 × N_RANKS	in	the 2 PRE winners / rank (pick-2-pre off p_bit)

Global / cross-rank legality (for the L3 CAS pick + packing)

signal	width	dir	meaning
can_cas_dq	1	in	DQ bus free (dqFree=BL/2); only <i>one</i> CAS may win the bundle
can_wr / can_rd	1 ea	in	R<->W turnaround gap (tWTR/tRTW) on a direction flip
last_cas_dir/rank/bg	few	in	bus state -> cross-rank tRTRS, diff-BG packing, dir bias
batch_dir[r]	1 × N_RANKS	in	WLR R/W batch direction (rank bias at L3)
gear	2	in	1/2/4 -> number of phase slots to fill

Maintenance flags (ME -> scheduler, ranked with demand)

signal	width	dir	meaning
refsb_rdy / ref_force	1 ea	in	refresh owed (debt) / forced (&debt==7)
rfm_rdy / rfm_force	1 ea	in	rowhammer owed (RAA>=IMT) / hard (>=MMT)
zq_rdy	1	in	ZQ calibration due
mrr_rdy	1	in	MR-read pending (temp poll / read MR) – uses DQ
mrw_rdy	1	in	MR-write pending (config / freq-change)
cas_complete[b]	1 × N_BANKS	in	bank drained + precharged (state==IDLE) – issue gate
maint_cmd_type	5	in	MAINT_PRIO winner opcode (REF/RFM/ZQ/MRW/MRR)
init_done	1	in	boot done (else DFI mux on init, scheduler gated)
dfi_alert_n / dfi_error	1 ea	in	PHY alert / error

2 OUTPUTS (from the scheduler)

Bundle -> cmd_phaser

signal	width	dir	meaning
CMD_0..3_PACKET (vld + record)	~49×4	out	{valid, cmd_type, rw, rank, bg, bank, addr, ap, bl, rob_index, sram_addr} per phase slot (1/2/4 used by gear)
two_ui / repeat	1	out	side wire -> DFI driver (1-UI vs 2-UI; @1:1 hold)

Grants / commit (back to cells + scoreboard)

signal	width	dir	meaning
grant_act_oh / grant_cas_oh	1 ea × N_BANKS	out	the chosen ACT/CAS winner one-hot; pop-on-CAS retires packet
grant_pre_oh[0..1]	1×2 × N_BANKS	out	the chosen PRE winners one-hot
fsm_commit (issue-edge)	pulse × class	out	sole commit: updates BANK_FSM + fires the timestamp writeback
cmd_issued{r,bg,bank,cmd}	bus	out	drives next_* writeback (WE + const-router select)

Maintenance handshake (to ME)

signal	width	dir	meaning
maint_ack	1	out	the maint cmd (cmd_type) issued -> ME clears debt/RAA, advances ba

Upstream (feeds the winners — NOT scheduler I/O)

The per-rank arbiters are a SEPARATE stage before the scheduler. Their I/O: **in** = PRE_BANK_ROW/COL_PATHS×32 (per-bank need/can_act/cas/pre, is_hit, age, open_row, col/ptr/rob), p_bit (PRE-elig vector), rank-scope gates can_act_rank_s/can_cas_rank_s, BG gates. **out** = act_win/cas_win/pre_win (the scheduler's inputs above). So can_*, p_bit, weights (CAS_W/ACT_W/ PRE_W) are consumed HERE, not in the scheduler bar.

Scheduler bar's job: take the per-rank winners + maint flags, apply the cross-rank / global constraints (one CAS via dqFree, turnaround, tRTRS, dir bias), rank maint WITH demand (MAINT_PRIO), and PACK the CMD_0..3 bundle for the phaser — plus feed back grant_oh (pop-on-CAS) and the fsm_commit writeback edge. It never rejects: legality was folded into the winners' rdy bits upstream.